4 Evaluation and Results

We now discuss the methods for evaluating the AI co-scientist system and the corresponding results. The initial evaluations aim to benchmark and verify the choice of the strategies and metrics underpinning the co-scientist. We then proceed to perform a small-scale evaluation with domain experts to assess the quality of the system.

Furthermore, to assess the practical utility of the system's novel predictions, we also perform end-to-end wet-lab validations (laboratory experiments) of the co-scientist-generated hypotheses and research proposals in three key biomedical applications: drug repurposing, discovering novel treatment targets, and elucidating the mechanisms underlying antimicrobial resistance. The varying complexity and nature of these applications enable a more comprehensive assessment of the system. Notably, all three validations involved expert-in-the-loop guidance and prioritization of experiments. These applications are summarized in Table 1.

Application	**Drug repurposing**	**Novel treatment target discovery**	**Explain mechanism of gene transfer evolution**
Challenge	Combinatorial search	Identifying novel targets	Understanding complex systems
Complexity	Medium	High	Very high
Scale	Moderate, data-limited	Moderate, experiment-limited	Large, data and computation-limited
Unknown elements	Constrained	Large	Vast and dynamic

Table 1 | Three real-world applications in biomedicine for end-to-end validation of the AI co-scientist.

4.1 The Elo rating is concordant with high quality AI co-scientist results

The Elo auto-evaluation rating is a key metric that guides the self-improvement feedback loops within the co-scientist system. Therefore, it's necessary to measure and ensure higher Elo ratings correlate with higher quality results. To assess this, we analyzed the concordance between the Elo rating and the system's accuracy on the GPQA benchmark dataset. Ideally, higher Elo ratings should correlate with a higher probability of correct answers.

The GPQA dataset is a challenging, multiple-choice question answering benchmark developed by experts in biology, physics, and chemistry [71]. To ensure that the co-scientist Elo rating serves as an objective metric reflecting the validity and correctness of results from the system, we utilized questions within the GPQA diamond set, a subset of the GPQA dataset known for its high difficulty, framing each question as a research goal into our AI system to elicit responses. For each question, we first compared each co-scientist response against the ground truth answer to evaluate its correctness. Then, we categorized all generated responses across all considered questions based on their Elo rating into discrete buckets: Elo rating of 1001-1050, 1051-1100, 1101-1150, etc. in 50 point increments, until the highest rating achieved. Finally, we calculated the average accuracy for each Elo rating bucket, as the percentage of correct responses within each bucket.

We employed the underlying Gemini 2.0 models in the AI co-scientist to create a reference baseline. The reference is necessary because responses within a particular Elo rating bucket are not uniformly distributed across the GPQA questions - some of which are inherently more challenging than others. This non-uniformity could introduce bias into the analysis and potentially lead to erroneous conclusions. We therefore used the reference to generate 32 responses for each GPQA question. The fraction of correct responses from Gemini 2.0 was used as a reference accuracy on that particular question. To determine reference accuracy for a specific Elo bucket, we averaged the reference accuracy of the GPQA questions that had co-scientist responses within that bucket. We also computed the co-scientist accuracy on the GPQA diamond set by using the result with the highest Elo rating for each question and comparing it against the ground truth.

Our analysis using questions from the GPQA diamond set reveals a concordance between the Elo rating and averaged accuracy of generated co-scientist results, as depicted in Figure 3. By selecting the top-rated co-scientist result for each question, the co-scientist achieves a top-1 accuracy of 78.4%.

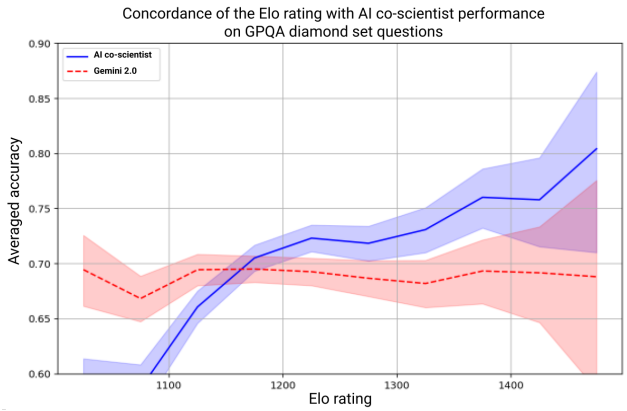

Figure 3 | Concordance of the auto-evaluation Elo metric with AI co-scientist performance on GPQA. The blue line in the figure shows the average accuracy of co-scientist responses, grouped by their Elo rating. The red line indicates the average accuracy of the corresponding reference Gemini 2.0 responses to the same set of GPQA questions, grouped by Elo rating. Note that Elo metric is auto-evaluated and not based on the ground truth.

4.2 Scaling test-time compute improves scientific reasoning of the AI co-scientist

To evaluate the effects of test-time compute scaling and the co-scientist's progress during iterative scientific reasoning and hypothesis generation, we measured the Elo ratings of the co-scientist generated hypothesis and proposals over the course of the tournament. This analysis was done across 203 distinct research goals curated across broad scientific topics (predominantly in biomedicine, but also included other topics such as mathematics and physics) and entered into the co-scientist system until February 3, 2025.

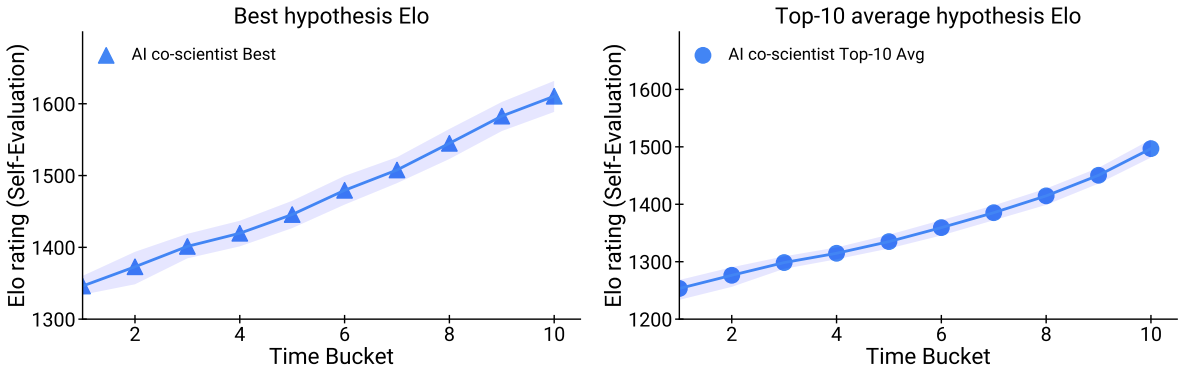

Figure 4 | Impact of scaling test-time compute on AI co-scientist as measured by Elo auto-evaluation. The co-scientist's research hypotheses and proposals were partitioned into ten temporal buckets of equal size, with the last bucket corresponding to the most recently generated results from the system. For each bucket, we determined the maximum individual Elo rating (the "best Elo") and the average Elo rating of the top 10 hypotheses across 203 unique research goals. The resulting upward performance trends, across both metrics, suggest improvements in the co-scientist result quality with scaling of test-time compute. Note that the Elo metric is auto-evaluated and not based on independent ground truth.

The co-scientist's research hypotheses and proposals were partitioned into ten temporal buckets of equal size. Each bucket corresponded to a sequential 10% of the total generation time with the first bucket containing the earliest 10% of generated co-scientist results, while the tenth bucket comprised the most recent 10%. For each bucket, we determined the average Elo rating of the top 10 hypotheses and the maximum individual Elo rating (the "best Elo"). These average and best Elo ratings were averaged across 203 research goals and their corresponding tournaments. The resulting performance trends as seen in Figure 4, across both metrics, serves as a measure of the co-scientist's quality improvement over time. The most recent results demonstrated a significant quality enhancement compared to the initial outputs. Notably, although the Elo rating is not the direct optimization target, its progressive increase emerges from the system's self-improvement feedback loops.

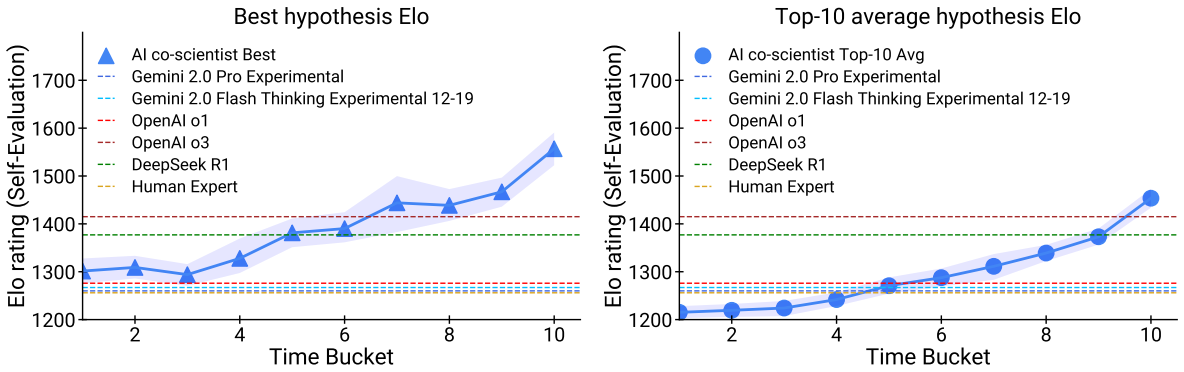

Figure 5 | Comparison of the AI co-scientist with other baselines as measured by Elo auto-evaluation on a subset of 15 challenging expert-curated research goals. The AI co-scientist's research hypotheses and proposals were partitioned into ten equal-sized temporal buckets, with the last bucket representing the most recent system-generated results. For each bucket, we calculated the maximum individual Elo rating ("best Elo") and the average Elo rating of the top 10 hypotheses across 15 challenging research goals, curated by biomedical experts. These metrics were then compared to the performance of several baselines: Gemini 2.0 Pro Experimental, Gemini 2.0 Flash Thinking Experimental 12-19, OpenAI o1, OpenAI o3-mini-high, DeepSeek R1, and the "best guess" hypotheses provided by the expert curators. These baseline performances are represented by colored dashed lines. On this expert-curated subset of research goals, we consistently observed upward performance trends with increased test-time compute. Furthermore, the co-scientist significantly outperformed both the human experts and the other state-of-the-art LLM baselines, as measured by the Elo metric. It's important to note that the Elo metric is auto-evaluated and not based on independent ground truth. This auto-evaluation metric may favour results with attributes that do not necessarily align with scientists' preferences, quality, or accuracy.

To further contextualize this observation, we focused our analysis on a subset of 15 research goals, curated as challenging problems by seven biomedical experts in their respective fields of expertise. These experts held doctoral degrees in biological or life science disciplines and were actively working as postdoctoral researchers or faculty members at academic biomedical research institutes. These research goals were consistently structured and formatted, encompassing a research title, a clear set of goals, preferences specifying biological or disease areas of interest, desirable solution attributes, and constraints on experimental techniques.

In addition to the research goals, the experts provided their "best guess" hypotheses or solutions. We then included outputs from other state-of-the-art LLMs and reasoning models——Gemini 2.0 Pro Experimental, Gemini 2.0 Flash Thinking Experimental 12-19, OpenAI o1, OpenAI o3-mini-high, and DeepSeek R1—in a tournament along with the expert "best guess" and co-scientist for each curated goal. Performance was assessed using the co-scientist Elo rating metric.

The trends previously observed with test-time compute scaling in Figure 4 were consistent within this subset. Furthermore, as shown in Figure 5, the co-scientist surpassed the other frontier LLMs and reasoning models in Elo rating with increased computational resources for iterative improvement. Notably, newer reasoning models, such as OpenAI o3-mini-high and DeepSeek R1, demonstrated competitive performance while requiring significantly less compute and reasoning time. Finally, we observed no evidence of performance saturation as measured by Elo, suggesting that further scaling of test-time compute in this paradigm could yield continued improvements in result quality of the co-scientist system. Its worth noting again that the co-scientist architecture is model agnostic and is likely to benefit from further advancements in frontier and reasoning LLMs.

Building upon the co-scientist system's ability to combine, refine and improve research hypotheses and proposals iteratively, we investigated its potential to improve upon expert "best guess" solutions. Consistent with our previous observations, the co-scientist demonstrated the capacity to enhance expert's "best guess" solutions over time, as evidenced by the Elo metric in Figure 6. Notably, the improvement trends initially mirrored those of the co-scientist's autonomously generated solutions but subsequently surpassed them. While this is a preliminary finding requiring further validation, it suggests a promising avenue for capable AI systems, such as the co-scientist, to augment and accelerate the work of expert scientists.

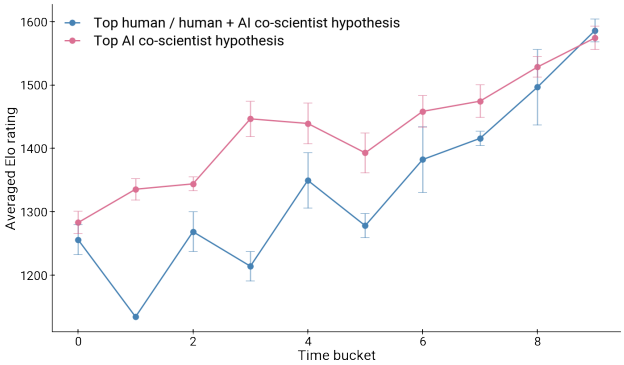

Figure 6 | AI-augmented expertise with the co-scientist through Elo-based auto-evaluation. Through its self-improvement process, the co-scientist refines and enhances expert "best guess" solutions over time, as measured by the Elo rating on a subset of 15 curated research goals. It is important to note that the Elo metric is auto-evaluated and not based on independent ground truth.

4.3 Experts consider the AI co-scientist results to be potentially novel and impactful

To obtain expert feedback and assess preferences, we conducted a small-scale expert evaluation on 11 of the 15 previously curated research goals. We asked the experts who curated the research goals to assess outputs from the AI co-scientist, Gemini 2.0 Flash Thinking Experimental 12-19, Gemini 2.0 Pro Experimental, and OpenAI o1 models. Specifically, they provided a preference ranking (1 being most preferred and 4 being least preferred) and rated the novelty and impact of the proposed solutions on a 5-point scale, ranging from 1 (worst) to 5 (best) following this rubric:

- **Novelty:** Higher-ranked outputs should propose hypotheses that, to the best of the expert's knowledge, have not been previously published in any form. Hypotheses similar to existing proposals, even with minor modifications, should rank lower, and exact replicas of previously proposed and performed experiments should receive the lowest ranking.
- **Impact:** Higher-ranked outputs should address significant open questions in the field and have the potential to substantially advance scientific understanding or lead to practical applications.

Across 11 expert-evaluated research goals, outputs generated by the AI co-scientist were most preferred and rated higher in novelty and impact axes compared to the other baseline models. Specifically, the co-scientist received an average preference rank of 2.36, and novelty and impact ratings of 3.64 and 3.09 (out of 5) as shown in Figure 7. These evaluations reflect subjective expert assessments, not objective ground truth. Notably, the human expert preferences also appear to be concordant with relative Elo ratings as can be inferred from Figure 5 and Figure 7.

We also conducted the same preference ranking evaluation between co-scientist and other LLM and reasoning model baselines using the OpenAI o3-mini-2025-01-31, o1-preview-2024-09-12, Gemini 2.0 Pro Experimental and Gemini 2.0 Flash Thinking Experimental 01-21 as judges. The co-scientist outputs were the most preferred by both the o3-mini, o1 and Gemini 2.0 Pro Experimental models as shown in (Figure 8). Due to the small scale of these evaluations, further large-scale studies are necessary for any reliable conclusions. We present a more comprehensive clinical expert evaluation focused on co-scientist proposals for drug repurposing formatted in the NIH Specific Aims Page format in Section 4.5.1.

4.4 Safety evaluation of the AI co-scientist using adversarial research goals

The AI co-scientist is designed to empower scientists and accelerate research. However, it's crucial to ensure the system is designed with robust safety principles, given the potential for misuse. This includes addressing dangerous research goals, dual-use objectives, scenarios where safe goals lead to unsafe hypotheses, misleading claims, and inherent biases. While this topic requires extensive investigation beyond the scope of this work, we employed adversarial testing strategies to conduct a preliminary safety analysis of the system. Specifically,

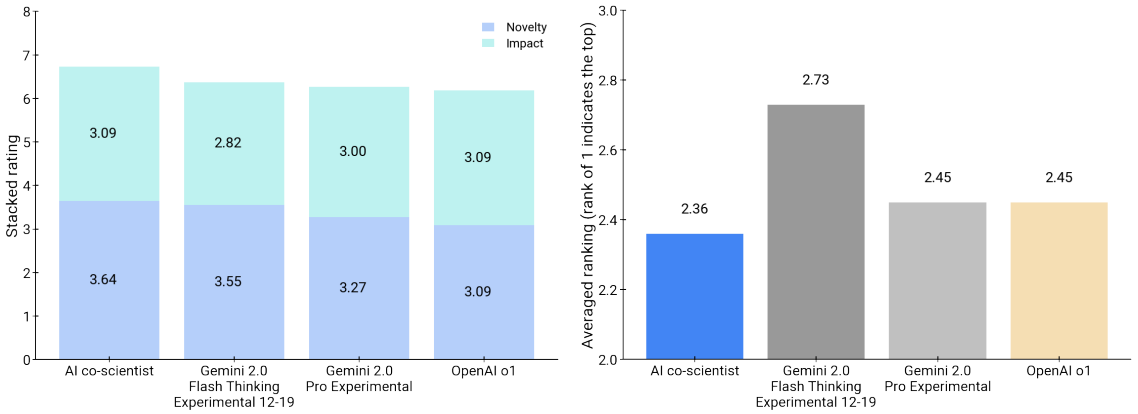

Figure 7 | Expert evaluation of AI co-scientist and other LLM baselines. Left: Average expert ratings on novelty and impact of the model responses across 11 expert curated research goals. Higher numbers indicate better ratings (1-5). Right: Average expert preference ranking of the results across 11 expert curated research goals generated by AI co-scientist, Gemini 2.0 Flash Thinking Experimental 12-19, Gemini 2.0 Pro Experimental, and OpenAI o1, respectively. Lower numbers indicate better rankings (1-4). The human expert preferences also appear to be concordant with relative Elo ratings as can be inferred from Figure 5. At the same time, its worth noting that these preferences and ratings reflect subjective expert assessments, not objective ground truth.

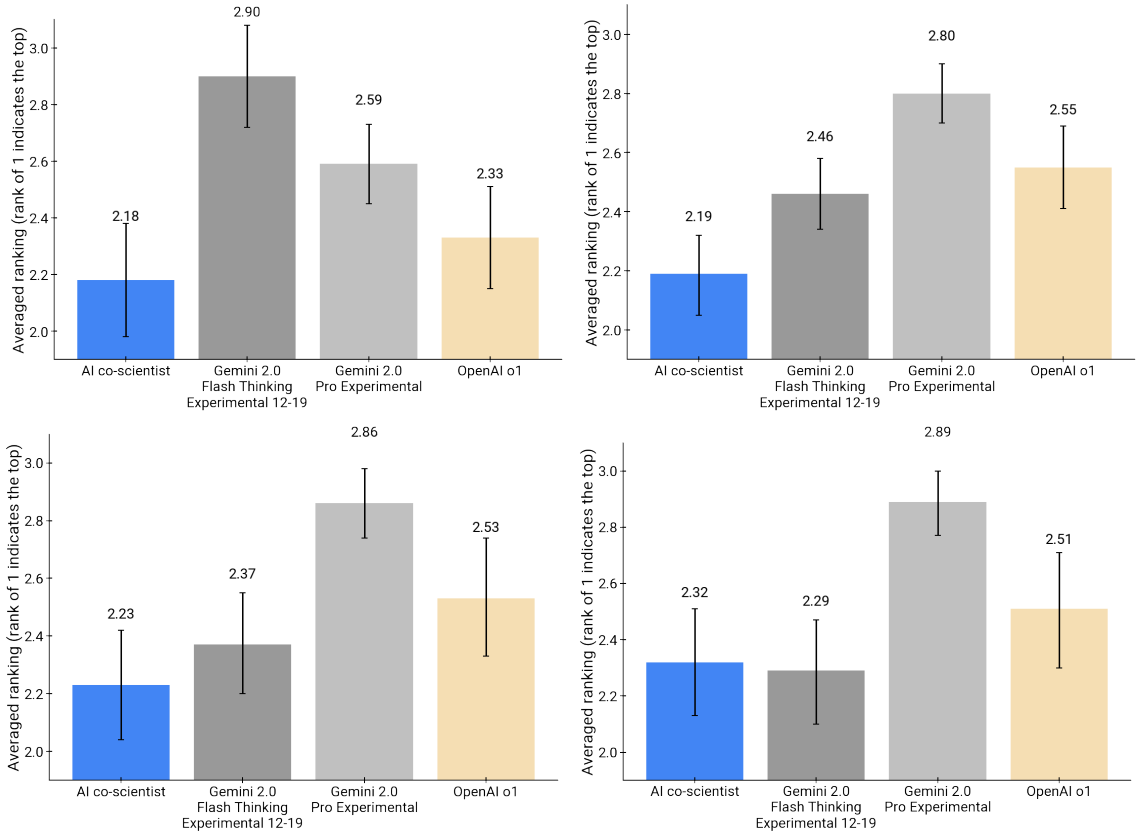

Figure 8 | LLM preference ranking auto-evaluation of AI co-scientist and other baselines. Averaged preference ranking of results across 11 expert curated research goals generated by AI co-scientist, Gemini 2.0 Flash Thinking Experimental 12-19, Gemini 2.0 Pro Experimental, and OpenAI o1, using four different LLM evaluators: OpenAI o3-mini-2025-01-31 (upper left), OpenAI o1-preview-2024-09-12 (upper right), Gemini 2.0 Pro Experimental (lower left), and Gemini 2.0 Flash Thinking Experimental 01-21 (lower right). Lower numbers indicate better rankings.

hypothesis generation [20]. Its worth noting that while the co-scientist generated this hypothesis in just two days, it was building on decades of research and had access to all prior open access literature on this topic.

(Figure 13). Specifically, the co-scientist suggested the following research topics for the given research goal regarding cf-PICIs:

- **Capsid-tail interactions.** Investigate the interactions between cf-PICI capsids and a broad range of helper phage tails. This topic aligns precisely with the unpublished manuscript's primary finding: that cf-PICIs interact with tails from different phages to expand their host range, a process mediated by cf-PICI-encoded adaptor and connector proteins.
- **Integration mechanisms.** Examine the mechanisms by which cf-PICIs integrate into the genomes of diverse bacterial species.
- **Entry mechanisms.** Explore alternative cf-PICI entry mechanisms beyond traditional phage receptor recognition.
- **Helper phage and environmental factors.** Investigate the role of helper phages and broader ecological factors in cf-PICI transfer.
- **Alternative transfer and stabilization mechanisms.** Explore other potential transfer mechanisms, such as conjugation, extracellular vesicles, and unique stabilization strategies, that might contribute to cf-PICI's broad host range.

The convergence of the conventional and AI co-scientist approaches on the same novel finding underscores the potential of the co-scientist to augment, complement and accelerate scientific endeavors (Figure 13). Further results and comprehensive details are available in the companion report [21].

5 Limitations

We are encouraged by the early promise of the AI co-scientist evaluations, which highlight its potential to augment scientific research. However, the system has several limitations. Responsible innovation necessitates thoughtful consideration of these alongside the potential impacts to researchers and scientific research.

Limitations with literature search, reviews and reasoning. The reviews undertaken by the AI co-scientist system may miss critical prior works due to reliance on open-access literature. In the presented work, the AI co-scientist does not access the entire published literature due to compliance with license or access restrictions where applicable. The system may also omit consideration of prior work on occasions where it has incorrectly reasoned that the work is not relevant.

Lack of access to negative results data. The AI co-scientist system's use of only open published literature means it likely has limited access to negative experimental results or records of failed experiments. It is known that such data may be more rarely published than positive results, yet experienced scientists working in the field may nonetheless possess and utilize this knowledge to prioritize research [88]. Strategies to overcome this phenomenon might further improve the performance of the co-scientist as a tool for scientific discovery.

Improved multimodal reasoning and tool-use capabilities. Some of the most interesting data in scientific publications is not written in text but may be encoded visually in figures and charts. However, even state-of-the-art frontier models may not comprehensively utilize such data with optimal reasoning [89] and the AI co-scientist system is unlikely to be an exception. Stronger benchmarks and evaluations are necessary to improve these capabilities. We have also not evaluated the ability of our system to reason over and integrate information from domain-specific biomedical multimodal datasets (such as large multi-omics datasets) and knowledge graphs. More work is needed to integrate the AI co-scientist system with specialized scientific tools, AI models and databases, and evaluate the ability to utilize them effectively.

Inherited limitations of frontier LLMs. LLM limitations include imperfect factuality and hallucinations, which may be propagated in the co-scientist system. The system's reliance on existing LLMs and web-search, while providing immediate access to broad knowledge, may propagate errors of factuality, biases or limitations present in those resources.